

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Wednesday 21 May 2025

Afternoon (Time: 40 minutes)

Paper reference **31617H/1P**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I

Physics

SECTION C: WAVES IN COMMUNICATION

You must have:
a calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
 - Section A: Structures and functions of cells and tissues (Biology)
 - Section B: Periodicity and properties of elements (Chemistry)
 - Section C: Waves in communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- The formulae sheet can be found at the back of this paper.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

- 1 Figure 1 shows a surface wave in a tank of water.

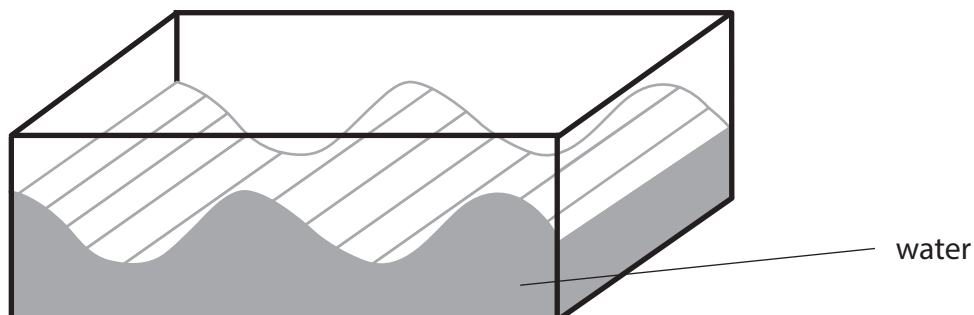


Figure 1

- (a) State the type of wave shown on the surface of the water in Figure 1.

(1)



(b) Figure 2 shows a graphical representation of the surface wave in the tank.

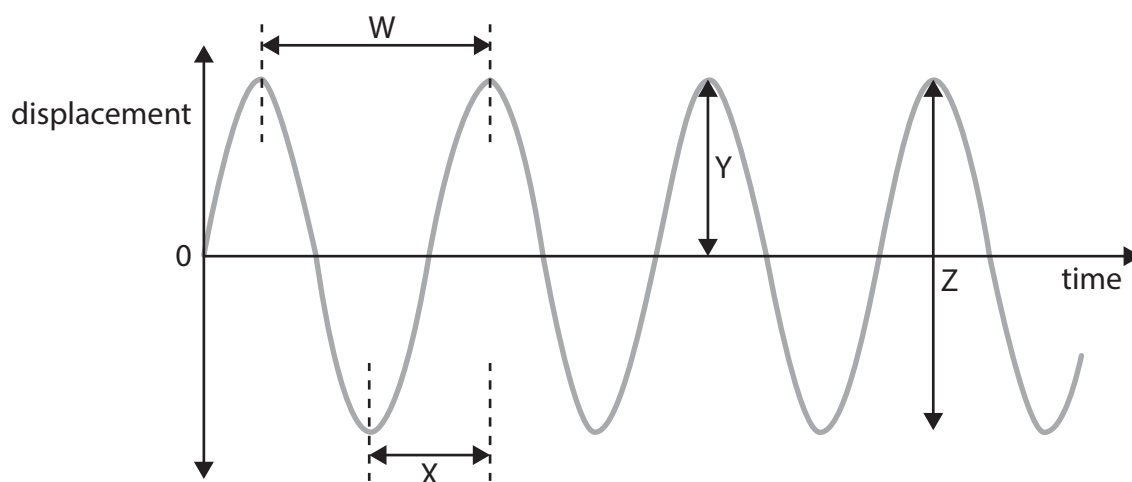


Figure 2

(i) Identify the arrow that shows the amplitude of the wave.

(1)

- ☐ **A** arrow W
- ☐ **B** arrow X
- ☐ **C** arrow Y
- ☐ **D** arrow Z

(ii) Identify the arrow that shows the time period of the wave.

(1)

- ☐ **A** arrow W
- ☐ **B** arrow X
- ☐ **C** arrow Y
- ☐ **D** arrow Z

(Total for Question 1 = 3 marks)



2 A TV station communicates with a satellite orbiting the Earth by sending a signal.

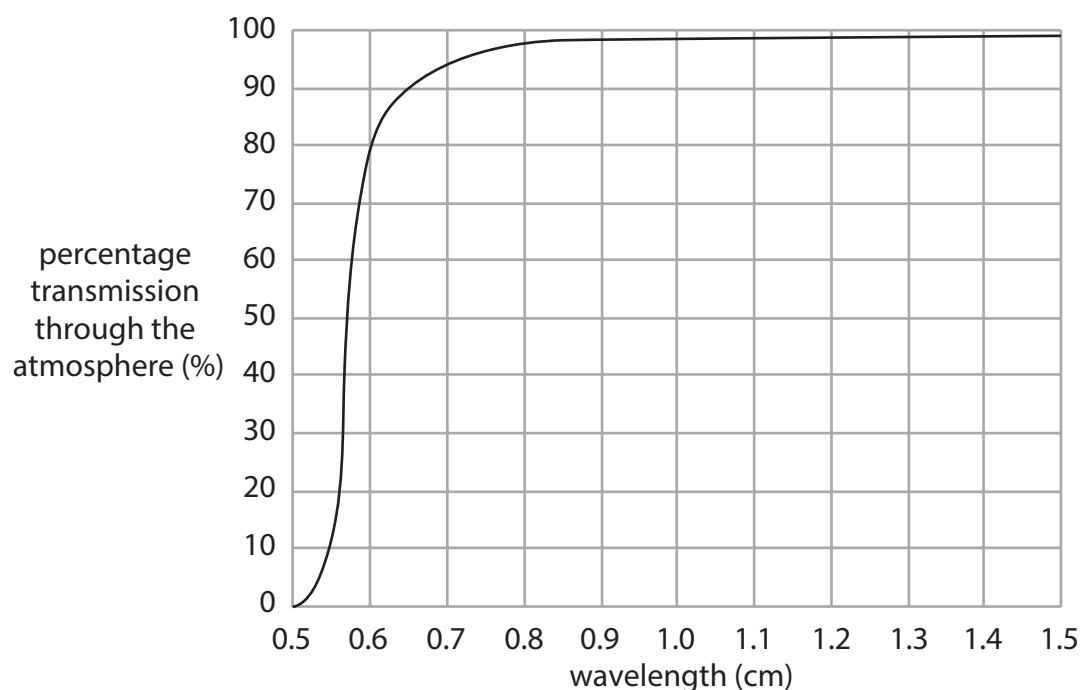
The signal consists of an encoded series of zeroes (0) and ones (1).

(a) Name the type of signal that consists of an encoded series of zeroes and ones.

(1)

The ability of a microwave to travel through the atmosphere depends on its wavelength.

Figure 3 shows the percentage transmission through the atmosphere for microwaves with wavelengths between 0.5 cm and 1.5 cm.



(Data Source: Schneider, Urban and Baron, 2009)

Figure 3

(b) Describe the trend shown in Figure 3.

(2)



Low frequency radio waves do not travel out of the atmosphere into space.

(c) Explain why low frequency radio waves do **not** travel out of the atmosphere.

(2)

(Total for Question 2 = 5 marks)



- 3 Figure 4 shows a ray of light travelling from the air into a solid glass cylinder.

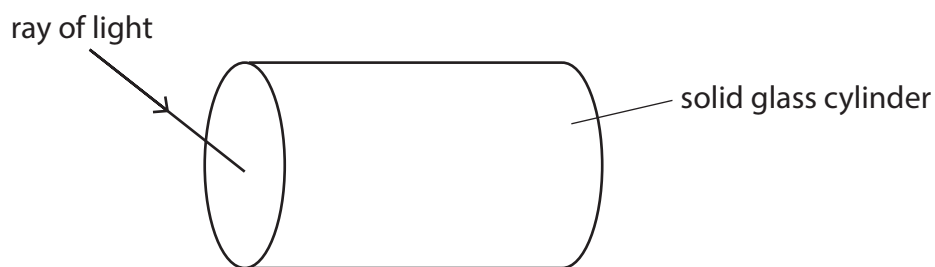


Figure 4

- (a) Describe what happens to the ray of light as it travels from the air into the glass cylinder.

(2)

.....

.....

.....

.....

The glass in the cylinder has a refractive index (n) of 1.6.

- (b) Calculate the speed of light in the glass (v).

(3)

The speed of light in air (c) is $3.0 \times 10^8 \text{ m s}^{-1}$.

Use the equation

$$n = \frac{c}{v}$$

Show your working.

speed of light in the glass = m s^{-1}



Solid glass cylinders can be used as optical fibres.

A medical endoscope has a thin tube containing two bundles of optical fibres.

The thin tube is inserted through a patient's mouth and into the stomach.

This enables an image of the stomach lining to be produced in the eyepiece of the endoscope.

- (c) Explain how an image of the stomach lining is produced in the eyepiece of the endoscope.

(4)

(Total for Question 3 = 9 marks)



4 Bluetooth® technology uses electromagnetic waves.

Figure 5 shows a Bluetooth® transmitter and two identical sensors, A and B.

The sensors measure the intensity of the wave emitted from the transmitter.

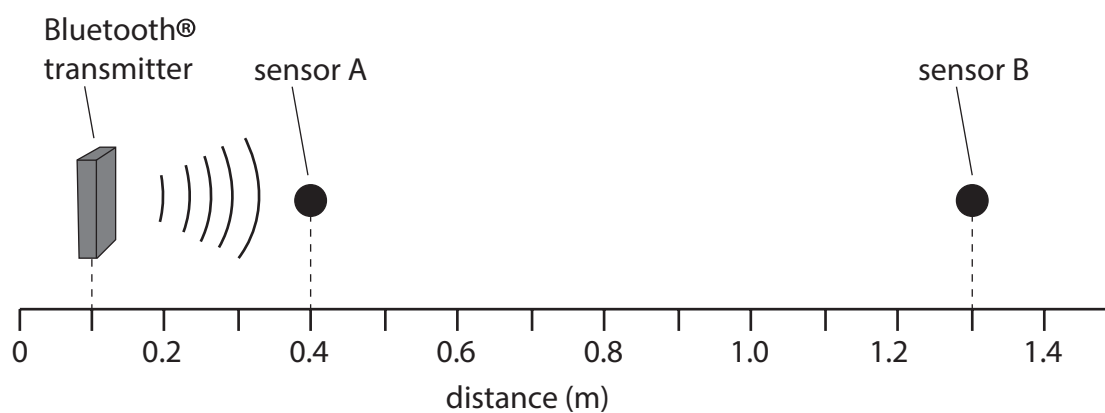


Figure 5

- (a) Explain why the intensity reading on sensor A is higher than the intensity reading on sensor B.

(2)

The intensity at sensor A (I_A) is 5.0 units.

(b) Calculate the intensity at sensor B (I_B).

(4)

Use the information in Figure 5.

Use the equation

$$\frac{I_A}{I_B} = \frac{(D_B)^2}{(D_A)^2}$$

I is the intensity of the signal at sensor A or B.

D is the distance between the transmitter and sensor A or B.

Show your working.

intensity at Sensor B (I_B) = units

(Total for Question 4 = 6 marks)



5 Some musical instruments use waves on strings to create sound.

The equation shows the speed of a wave on a string.

$$v = \sqrt{\frac{T}{\mu}}$$

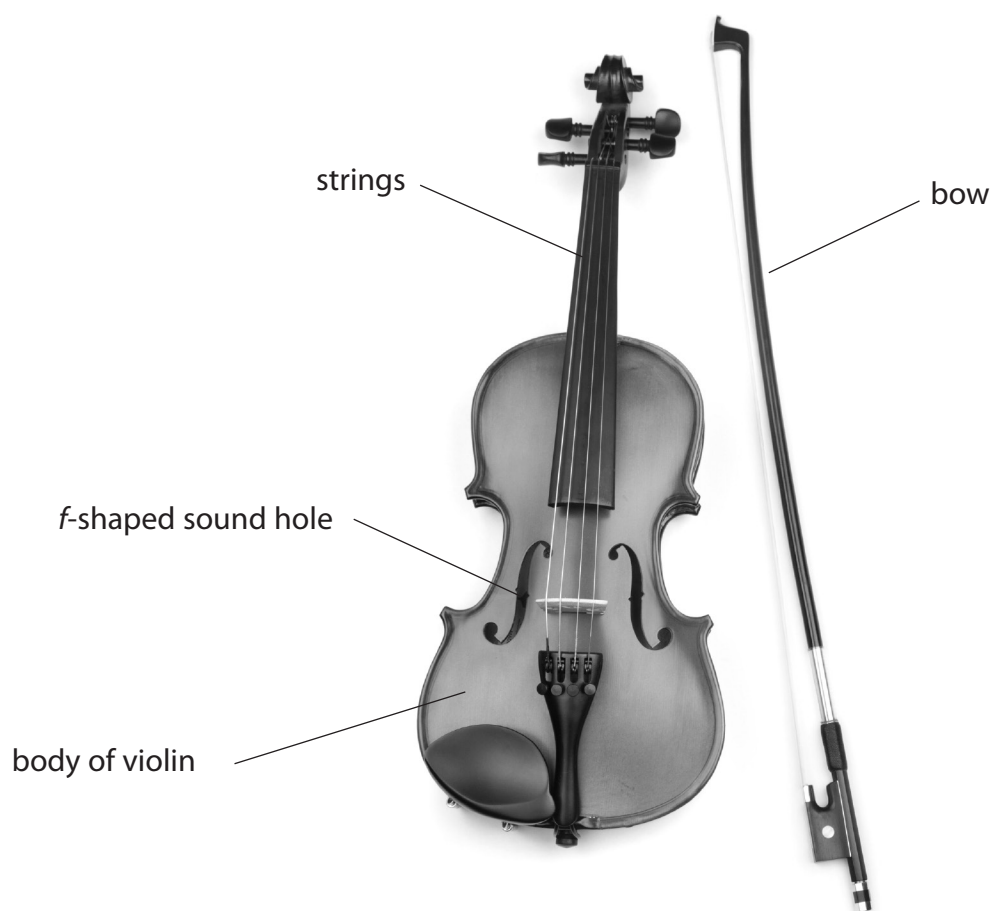
(a) Which quantity is represented by μ in the equation?

(1)

- ☐ A amplitude per unit length
- ☐ B force per unit length
- ☐ C frequency per unit length
- ☐ D mass per unit length

Figure 6 shows a violin and a bow.

A violin has four strings and a hollow wooden body with two *f*-shaped sound holes.



(Source: © PAL)

Figure 6



When the bow is pulled across the strings, the violin makes a sound.

- (b) Discuss how pulling the bow across the violin strings makes a sound that can be heard.

(6)

Handwriting practice area with 20 horizontal dotted lines.



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(Total for Question 5 = 7 marks)

TOTAL FOR PAPER = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS



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Formulae sheet

Wave speed

$$v = f\lambda$$

Speed of a transverse wave on a string

$$v = \sqrt{\frac{T}{\mu}}$$

Refractive index

$$n = \frac{c}{v} = \frac{\sin i}{\sin r}$$

Critical angle

$$\sin C = \frac{1}{n}$$

Inverse square law in relation to the intensity of a wave

$$I = \frac{k}{r^2}$$

$$\frac{I_1}{I_2} = \frac{(D_2)^2}{(D_1)^2}$$

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